Assignment 10

Aim - Implement a program to count number of leaf nodes, biggest and smallest number and height of a binary search tree.

Theory

A binary search tree is a tree in which the left node has the smaller number and right node has the larger number than its root.

A leaf node is a node where the left and right are both null, the tree ends at them.

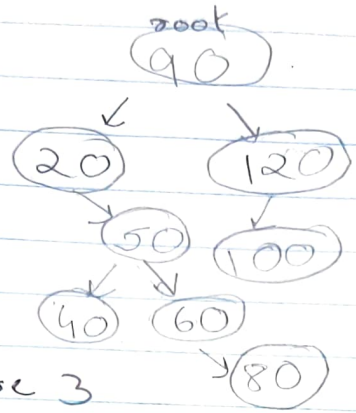
The smallest element is the leftmost leaf node because smaller values always on the left

The largeot element is the rightmost leaf node because largest values are alway on the right

Height of a atree is the number of edges or levels from root to deepest leaf.

THADOMAL SHAHANI
TSEC
ENGINEERING COLLEGE

e.g.



Sample data
90, 20, 50, 120, 100, 40, 60, 80

smallest → leftmost → 20
largest → Rightmost → 120
height → 3 (3 branches + root)

leaf nodes are 3
as they are at the ends :- 40, 80, 100

Conclusion

Binary search tree was successfully implemented. Functions were created to count leaf nodes, determine the ~~sma~~ smallest and largest elements, ~~calc~~ and calculate the height of the tree. The result verified that BST properties allow efficient searching and traversal operations.